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Diet of the Humboldt penguin (*Spheniscus humboldti*) in northern and southern Chile

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Abstract The diet of the Humboldt penguin (*Spheniscus humboldti*) was examined and compared in two colonies in Chile. Field work was conducted on Pan de Azúcar Island in northern Chile in the breeding season 1998/1999 and on the Puñihuil Islands in southern Chile over two successive breeding seasons during 1997/1998 and 1998/1999. Penguin diet was studied by stomach-pumping birds and analysed by species composition, size and mass of prey. Fish were the dominant prey item at both sites, the contribution of cephalopods and crustaceans varying between sites. The fish prey consisted predominantly of school fish, but there were clear latitudinal differences in fish prey taken. Penguins in the northern colony consumed primarily garfish (*Scombre-sox saurus*), while birds at the southern colony of Puñihuil fed primarily on anchovy (*Engraulis ringens*), Araucanian herring (*Strangomera bentincki*) and silverside (*Odontesthes regia*). The results showed significant differences in terms of numbers of fish taken between the two breeding seasons at Puñihuil. In 1997/1998 penguins consumed almost exclusively anchovy, while they fed primarily on silversides in the successive year. Almost all prey, except stomatopods, were characterised as being pelagic species that occur in relatively inshore water, consistent with the foraging behaviour of Humboldt penguins. The dependence of Humboldt penguins on

commercially exploited, schooling prey species makes the species particularly susceptible to changes in prey stocks, due to non-sustainable fisheries management.

Introduction

Investigations on seabirds have shown that food selection has an important influence on the biology of the birds: it affects bird activity, distribution, energetics, competition, breeding success and rate of survival (e.g. Furness and Monaghan 1987; Montevecchi et al. 1988; Carey 1996). Knowledge of feeding ecology and diet composition therefore is an important basis for understanding bird biology. Diet composition allows estimation of the impacts of changes in food availability, i.e. due to climatic changes or to anthropogenic influences.

Seabirds breeding along the coast of Chile encounter different feeding conditions, resulting from climatic differences between northern and southern Chile. This is also the case for the Humboldt penguin (*Spheniscus humboldti* Meyen, 1834), with a breeding range along the Pacific coast of South America from Isla Foca in the north (5°S) to the Puñihuil Islands in the south (41°S) (Williams 1995). Along the coastline the Humboldt penguin faces many different ecological and climatic conditions from dry coastal desert climate and nutrient-rich coastal upwelling regions in the north to lush temperate rainforests and poor coastal upwelling in the south (e.g. Lalli and Parsons 1997; Hennicke 2001). The coastline of Chile is influenced by the cold, nutrient-rich Humboldt Current, flowing northwards from Antarctica (Colling 2001), bringing waters with colder temperatures than typical for the prevailing latitudes (Dietrich et al. 1975). In the north of Chile the cold sea-surface temperatures are being intensified by the coastal upwelling that usually brings up cold, nutrient-rich waters from below the thermocline (Colling 2001). This, in turn, allows high primary productivity (Arntz and Fahrbach

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1991), resulting in high amounts of fish such as the plankton-eating anchovy (*Engraulis ringens*) (Lalli and Parsons 1997). The upwelling system of the Humboldt Current is regularly affected by recurrent periods of anomalous climatic conditions known as El Niño (ENSO: El Niño Southern Oscillation), where the nutrient-rich surface waters are being replaced by warm nutrient-poor waters (Colling 2001) and fish becomes unavailable to seabirds (e.g. Hays 1986; Boersma 1998).

Humboldt penguin populations have dropped dramatically (Hays 1986; Luna-Jorquera 1996a). The commercial fishery that occurs especially along coastal upwelling systems like those of South America and South Africa (Duffy et al. 1987a; Montevecchi et al. 1988) is potentially capable of harming penguin populations as in the case of the African and Magellanic penguins (Crawford and Shelton 1981, cited in Sclaro et al. 1999.). The commercial fishery is the biggest threat to Humboldt penguins nowadays (Ellis et al. 1998). Other factors that severely affect the populations are entanglement and drowning in nets (Simeone et al. 1999) and recurring food-shortage during El Niño events (e.g. Hays 1986). Humboldt penguins have recently been classified as vulnerable, the population remaining at low levels since the early 1980s (Luna-Jorquera and Culik 2000).

Although the Humboldt penguin is one of the main predators of the Humboldt Current, only little is known about its feeding preferences along the coast off Chile, with its climatic differences between the north and the south. Until now, the prey of Humboldt penguins has, to our knowledge, only been documented by Murphy (Lalli and Parsons 1936) and Wilson et al. (1989, 1995). The substantial population decline together with expanding human activities, including commercial fisheries, suggests that investigations on feeding preferences of the Humboldt penguin have long been overdue.

In the present study we investigated the diet of Humboldt penguins in two Chilean colonies and over two successive breeding seasons in one colony in order to examine annual and regional differences. Our aims here were to characterise the penguins diet, compare site- and year-specific variation and to assess implications for competition with commercial fisheries. Overall, the study shall help to better understand the role of the Humboldt penguin in the marine ecosystem.

Materials and methods

Study areas

Field work was conducted in two breeding colonies of the Humboldt penguin (*Spheniscus humboldti*) in Chile, which are 1,500 km apart: Pan de Azúcar Island (26°09'S; 70°41'W) in Pan de Azúcar National Park in northern Chile during the breeding season 1998/1999 and Puñihuil Islands in the south (41°55'S; 74°02'W)

during the breeding seasons 1997/1998 and 1998/1999. Pan de Azúcar Island is located about 1.2 km from the coast, at the edge of the Atacama Desert. The cold temperate surface water along the coast is maintained by the Humboldt Current, flowing northwards from Antarctica (Colling 2001). The island is about 2 km long and 1 km wide. Pan de Azúcar is one of the most important breeding colonies of Humboldt penguins in northern Chile (Luna-Jorquera 1996a), with 800 birds counted on the island in 1998/1999 (Hennicke, unpublished data). The Puñihuil Islands are located on the west coast of Chiloé Island in southern Chile. The islands are exposed to high precipitation and low temperatures all year round, with temperatures seldom exceeding 11°C during summer. As opposed to Pan de Azúcar, the area surrounding the Puñihuil Islands is not influenced by the upwelling ecosystem of the Humboldt Current (Lalli and Parsons 1997). Isla Chica, the smallest of the islands, is about 0.4 km offshore. The largest island, Isla Grande, is located about 0.7 km from the coast. The Puñihuil Islands have the only known mixed colony of Humboldt and Magellanic penguins (*Spheniscus magellanicus*) in the world (Simeone and Schlatter 1998), with 150 Humboldt and 500 Magellanic penguins counted in the year of study (Hennicke, unpublished data).

Collection of food samples

Stomach samples were obtained from birds returning to the colonies by the water offloading technique of Wilson (1984). In the breeding season 1997/1998 penguins were flushed two to five times. In 1998/1999 penguins were only flushed twice. During the breeding season 1997/1998 stomach samples on Puñihuil Island were collected as soon as penguins returned back from the sea. For this purpose each nest was individually checked every half hour. In 1998/1999 stomach samples in both colonies were obtained from penguins returning from a foraging trip, caught on their way back from the shore to the nests. Most of the penguins were breeding adults returning to their nests; some were moulted and no longer breeding.

In Pan de Azúcar field work was conducted in two intervals. The first sampling period was between 28 November and 2 December 1998. The second sampling period was 1 week later between 9 and 18 December 1998.

Analysis of food samples

The digested part of the stomach content was separated from the undigested, i.e. still identifiable material like whole fish, fish tails, fish heads, squid, or crustaceans. The digested chyme was weighed and immediately placed in 95% isopropanol for storage and transport to the laboratory for later analysis. The undigested mate-

rial was identified directly whenever possible and then dried and stored for transport. Analyses were conducted at the Institut für Meereskunde, Kiel, Germany. Each sample was sorted into fish, cephalopod and crustacean components. For analysing the digested material, the chyme was drained through a 90-µm sieve and then examined for the presence of any diagnostic prey remains like otoliths, vertebrae, cephalopod beaks, or crustacean carapaces. Diet composition from each site was assessed from complete samples in four ways:

1. frequency of occurrence,
2. number of individuals,
3. wet mass of prey, and
4. length of prey.

Fish

Loose otoliths were collected from stomach samples, and those still enclosed in fish crania were removed. Otoliths were identified by direct comparison with a reference collection compiled for this purpose by Dr. C. Moreno, Universidad Austral de Chile, Valdivia, Chile, and by using the identification key from Gellona et al. (1995). The identification was verified at Universidad de Concepción, Chile, by Dr. M. Araya. Because only few otoliths were found, fish species were also determined by examining vertebrae. Loose vertebrae were collected from the samples, and those still enclosed in fish bodies were removed. The vertebrae were identified by direct comparison with a reference collection. For this purpose samples of anchovy (*Engraulis ringens*), pilchard (*Strangomera bentincki*) and silverside (*Odontesthes regia*) were caught, weighed and measured in Chile and sent to Germany, where the vertebral columns were removed from the fish samples.

Other species were identified with the key by Gellona et al. (1995). For identification purposes, the caudal vertebrae were used, because they show less variations than abdominal vertebrae and are therefore easier to compare. Furthermore caudal vertebrae are very robust and more resistant to digestion than other vertebrae (Watt 1997).

Due to the low number of otoliths found in the samples, the first caudal vertebrae were used to estimate the number of fish consumed. When no first vertebrae were present, the anterior or, when no anterior caudal vertebrae were present, the posterior caudal vertebrae were counted. The total number was divided by the number of anterior or posterior caudal vertebrae, which was counted on average in fresh fish samples to obtain the total number of fish in the sample.

Loose otoliths were paired by species and size together with the otoliths of intact heads enclosing paired otoliths. Otoliths were considered to represent an additional fish, when representing a species that was not already represented by the vertebrae.

The length of otoliths and vertebrae were determined using a calibrated reticule in a microscope. Regressions relating vertebra length or otolith length, respectively, to fish length were established to allow estimation of the original fish size. Regressions were obtained from the biometric data derived from the otolith and vertebra collection from fresh fish samples of *Engraulis ringens*, *Strangomera bentincki* and *Odontesthes regia*. Regressions relating fish length and body mass were obtained from direct measurements of fresh fish. Lengths of fish specimens other than the three species mentioned above were calculated using the proportional method (Watt 1997):

$$l_1 = q_1 \times l_2 / q_2 \quad (1)$$

where l_1 is the length of the unknown fish from the sample (mm), q_1 is the length of vertebrae from the sample (mm), l_2 is the length of the reference fish (mm) and q_2 is the length of vertebrae of the reference fish (mm).

This method requires knowledge of the relationship between fish length and length of the first caudal vertebrae (when not present, other caudal vertebrae) from at least one specimen of the identified species. Regressions relating fish length and body mass were obtained in this case from the literature or from the internet (<http://www.fishbase.org>).

Cephalopods

All loose cephalopod beaks were removed from the stomach contents, sorted into upper and lower beaks, counted and stored in alcohol. Intact buccal masses were counted separately before extraction of lower beaks. Intact specimens and beaks were identified from the literature with the keys by Clarke (1986), Rocha (1997), Wolff (1982, 1984) and where possible by comparison with reference material obtained from Dr. U. Piatkowski, Institut für Meereskunde, Kiel, Germany. The total number of cephalopods was estimated from the combined number of loose lower beaks and intact buccal masses. Regressions were obtained from the literature and applied to all intact beaks in order to estimate the size of the cephalopods. For *Loligo gahi*:

$$\ln M \text{ (g)} = 2.25 + 2.39 \ln \text{LRL (mm)} \quad (2)$$

and

$$\ln \text{ML (mm)} = 4.23 + 1.01 \ln \text{LRL (mm)} \quad (3)$$

both equations according to Piatkowski et al. (2001); and for *Dosidiscus gigas*:

$$\ln M \text{ (g)} = 7.4 + 2.48 \ln \text{LRL (cm)} \quad (4)$$

and

$$\text{ML (mm)} = 44.2 + 357.9 \text{LRL (cm)} \quad (5)$$

both equations according to Wolff (1984), where LRL is the lower rostral length, ML is the dorsal mantle length and M is the body mass.

Crustaceans

All crustaceans were counted and identified. Only stomatopods and isopods were found. The stomatopods were identified to the lowest taxonomic level possible from literature with the keys by Manning (1968) and the Food and Agriculture Organization (FAO 1993). The identification of the isopods to species or genus level was not possible, due to rate of digestion and small size. The lengths of all intact stomatopod specimens were measured from the tip of the rostrum to the tip of the uropod. A regression from a related species (*Oratosquilla nepe*) of the same family, obtained from the literature, was used to estimate the body mass of the stomatopods:

$$\log M = -4.9665 + 2.9661 \log L \quad (6)$$

according to James and Thirumilu (1993), where L is the length (mm) and M is the body mass (g).

Statistics

All data were statistically treated using SPSS 10.0 for Windows. Data were tested for normality by using Kolmogorov–Smirnov’s adaptation test (Lamprecht 1992). Due to the non-normal distributed data only non-parametric tests were used. Two independent random samples were compared by Mann–Whitney U -test. Frequencies were compared by χ^2 -test. When N was too small or the expectation values were < 5.0 , relative frequencies were tested by Fisher’s test. All Fisher tests were two-tailed. In all tests, $P < 0.05$ was accepted as indicating statistical significance.

Results

During the breeding season 1997/1998, 11 stomach samples were collected on the Puñihuil Islands; in the following season, 1998/1999, the stomachs of 8 penguins could be flushed. At Pan de Azúcar, 22 stomach samples were collected in 1998/1999. During the first sampling

period the stomachs of 7 penguins were flushed; 15 samples were taken during the second sampling period.

Frequency of occurrence

At both sites, fish was the predominant prey item in terms of frequency of occurrence. At Pan de Azúcar and Puñihuil it appeared in every sample (Table 1). Cephalopods were also found at both sites. This prey taxon occurred in approximately 55% of penguin stomach samples in Puñihuil in the breeding season 1997/1998 and in 63% of the samples in 1998/1999. In Pan de Azúcar cephalopods occurred in 41% of the samples. Penguins at Puñihuil rarely consumed crustaceans in both seasons, while this prey taxon occurred in 45% of penguin stomach samples from Pan de Azúcar.

Fish

Overall, Humboldt penguins fed predominantly on pelagic prey species, particularly those occurring in aggregations. There were well-defined, area-specific dietary differences.

At Puñihuil penguins fed in both years on anchovy (*Engraulis rigens*), Araucanian herring (*Strangomera bentincki*), silverside (*Odontesthes regia*), common hake (*Merluccius gayi*) and Inca scad (*Trachurus murphyi*). The compositions of species were the same in both breeding seasons, but there were significant differences in the occurrence of silversides in the samples. In 1997/1998 penguins rarely consumed silversides (9%), while this species occurred in approximately 88% of penguins stomach samples in the successive breeding season ($P = 0.002$, Fisher’s test) (Table 2).

At Pan de Azúcar penguins fed on anchovy and Inca scad and additionally on garfish (*Scomberesox saurus*) and South American pilchard (*Sardinops sagax*). No Araucanian herring, silverside, or common hake were found.

Garfish occurred in every penguin stomach sample from Pan de Azúcar during both sampling periods. Anchovy was the second most important fish species and occurred in $> 40\%$ of penguin stomach samples during both sampling periods (Table 3).

Table 1 Frequency of occurrence (%) of fish, cephalopods and crustaceans in the stomach contents of Humboldt penguins (*Spheniscus humboldti*) in Chile

	Puñihuil			Pan de Azúcar	
	1997/1998	1998/1999	All samples	First sampling period	Second sampling period
Samples (n)	11	8	22	7	15
Fish	100	100	100	100	100
Cephalopods	55	62.5	41	71	18
Crustaceans	18	25	45	71	23

Table 2 Frequency of occurrence (%) of the different fish species in the stomach contents of Humboldt penguins (*Spheniscus humboldti*) at the Puñihuil Islands

	1997/1998	1998/1999
Anchovy (<i>Engraulis ringens</i>)	91	62.5
Araucanian herring (<i>Strangomera bentincki</i>)	91	50
Silverside (<i>Odontesthes regia</i>)	9	87.5
Common hake (<i>Merluccius gayi</i>)	18	12.5
Inca scad (<i>Trachurus murphyi</i>)	9	25

Cephalopods

All intact cephalopod beaks from the stomach samples of Puñihuil were identified as *Loligo gahi*. At Pan de Azúcar two species were identified from all intact cephalopod beaks in the samples; *Dosidiscus gigas* occurred in 32% of penguin stomach samples, while *L. gahi* occurred in approximately 5% of the samples. Marked changes in the cephalopod diet were measured between the two sampling periods at Pan de Azúcar. During the first sampling period cephalopods occurred in 71% of penguin stomach samples, while this prey species occurred only in approximately 18% of the samples 1 week later (Table 1).

Crustaceans

At Puñihuil nearly all crustacean carapaces found in the samples were identified as isopods. The degree of digestion combined with the small size prevented further identification. At Puñihuil isopods only occurred in 2 of the 11 samples (18%) in the first year 1997/1998 and in 2 of the 8 samples (25%) in the following year (Table 1). Only one stomatopod was found in one sample during the season 1998/1999. It was identified to the family level as a Lysiosquillidae.

At Pan de Azúcar stomatopods occurred in 3 of the 22 samples, all taken during the first sampling period. All stomatopods belonged to the family Squillidae.

Table 3 Frequency of occurrence (%) of the different fish species in the stomach contents of Humboldt penguins (*Spheniscus humboldti*) at Pan de Azúcar over all samples and during both sampling periods

	All samples	First sampling period	Second sampling period
Garfish (<i>Scomberesox saurus scombroides</i>)	100	100	100
Anchovy (<i>Engraulis ringens</i>)	45	43	47
Inca scad (<i>Trachurus murphyi</i>)	14	14	13
South American pilchard (<i>Sardinops sagax</i>)	9	14	7

Some were identified to the genus level as *Pterygosquillidae* or as the species *Pterygosquillidae gracilipes*. Isopods occurred in 71% of the samples at Pan de Azúcar during the first sampling period, while this prey species occurred only in approximately 23% of the samples during the second sampling period.

Composition by numbers

At Puñihuil a total of 195 prey items were analysed in 1997/1998, and 145 prey items, in the following season (1998/1999). A total of 412 prey items were examined at Pan de Azúcar. At both sites penguins fed predominantly on fish in terms of numbers.

At Puñihuil, in 1997/1998, fish accounted for 93% of all items eaten, followed by cephalopods (6%) and crustaceans (1%), respectively. In 1998/1999 fish accounted for almost 88% of all items consumed. At Pan de Azúcar fish constituted >75% of the items eaten, followed by crustaceans (17%) and cephalopods (8%).

Marked changes in the diet were recorded between the two sampling periods at Pan de Azúcar. The proportion of fish consumed was lower during the first sampling period than in the second. From the total of 164 prey items counted, 79 were identified as fish in the first season (48%), whereas in the second season, from a total of 248 prey items, 229 were analysed as fish (92%) ($P < 0.001$, χ^2 -test). Crustaceans accounted for almost 37% ($n = 63$) of all items consumed during the first sampling period, while penguins rarely consumed crustaceans during the second sampling period (3%) ($n = 8$) ($P < 0.001$, χ^2 -test). Cephalopods constituted the remaining 13% ($n = 22$) of the diet in the first sampling period and 4% ($n = 11$) during the second sampling period, respectively. This difference was significant ($P < 0.01$, χ^2 -test).

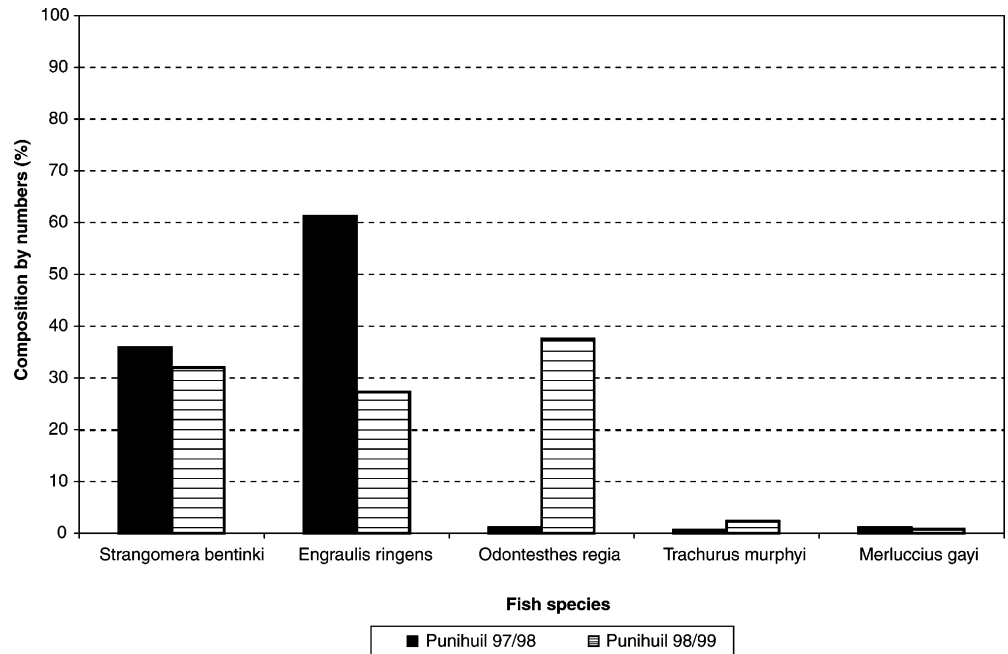
Fish

At Puñihuil a total of 309 fish were counted (181 in 1997/1998 and 128 in 1998/1999). In Pan de Azúcar a total of 308 fish were counted.

There was a significant seasonal difference in the numbers of anchovies ($P < 0.001$, χ^2 -test) and silversides ($P < 0.001$, χ^2 -test) eaten by the penguins at Puñihuil (Fig. 1). In 1997/1998 penguins fed almost exclusively on anchovy (*E. rigens*). From the total of 181 fish consumed 111 were identified as anchovy, which constituted 61% of the fish diet in terms of numbers. In 1998/1999, from the total of 128 fish counted, anchovy made up only 27% ($n = 35$), while silversides (*O. regia*) accounted for almost 37% ($n = 48$) of the fish consumed.

At Pan de Azúcar the penguins fed almost only (> 70%) on garfish (*S. saurus*). Anchovy was the second most abundant fish species, accounting for 22% of all 308 fish consumed. The other two fish species contributed only to a smaller degree to the diet with 5%

Fig. 1 Composition by numbers (%) of the fish component of the diet of Humboldt penguins (*Spheniscus humboldti*) at Puñihuil in 1997/1998 and 1998/1999



analysed as Inca scad (*T. murphyi*) and 2% as pilchard (*S. sagax*). There were significant differences between the two sampling periods in the numbers of garfish ($P < 0.001$, χ^2 -test) and anchovy ($P < 0.001$, χ^2 -test) eaten (Fig. 2). During the first period, from the total of the 79 fish counted approximately half of the fish were anchovies (46%), garfish being the next most abundant group of fish, with 30 items consumed (38%).

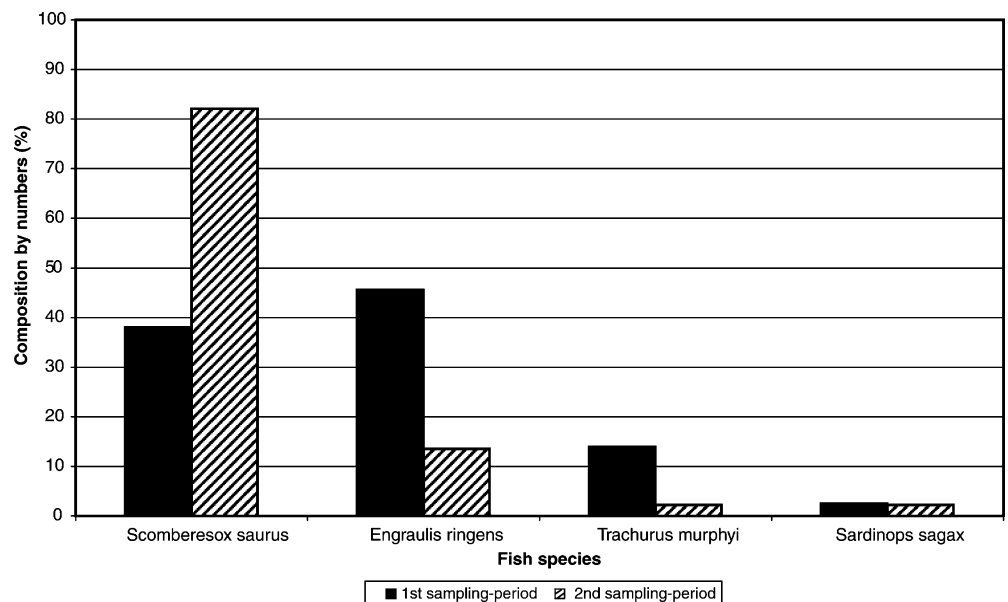
During the second sampling period garfish accounted for 82% ($n = 188$) of all 229 fish consumed, followed by anchovy accounting for 14% ($n = 31$). The contribution of *T. murphyi* varied significantly ($P < 0.001$, χ^2 -test) between the samplings (Fig. 2). From all 248 prey items

eaten, including cephalopods and crustaceans during the second sampling period, the proportion of garfish consumed was more than four times higher (76%, $n = 188$) than in the first sampling period (18%, $n = 30$), where a total of 164 prey items were identified ($P < 0.001$, χ^2 -test).

Cephalopods

At Puñihuil, numbers of cephalopod beaks, all identified as *L. gahi*, were almost equal during both breeding seasons. From the total of 195 prey items identified in the first season, 12 were identified as cephalopods. In

Fig. 2 Composition by numbers (%) of the fish component of the diet of Humboldt penguins (*Spheniscus humboldti*) at Pan de Azúcar in the samples during the first sampling period and during the second sampling period



1998, 13 of the 145 prey items consumed were cephalopods. At Pan de Azúcar, of the 33 cephalopods counted in all samples, only 15 beaks were identified as *D. gigas* and 4 as *L. gahi*. Identification of the beaks was difficult due to advanced digestion or because the beaks were broken. More cephalopods were consumed during the first sampling period ($P < 0.01$, χ^2 -test).

Crustaceans

The contribution of the two groups of crustaceans, isopods and stomatopods, to stomach contents was different between sites, as well as between the two sampling periods at Pan de Azúcar. During both seasons at Puñihuil and during the second sampling period at Pan de Azúcar, crustaceans constituted $< 5\%$ of numbers of all prey items consumed and these were primarily isopods. At Pan de Azúcar stomatopods occurred in large numbers during the first sampling period and constituted 31% of the number of all prey items consumed. Isopods, which occurred in $> 70\%$ of the samples during the first sampling, constituted only 7% of all items eaten in terms of numbers.

Composition by wet mass and length of prey

Fish predominated in the diet by wet mass at both sites followed by cephalopods and crustaceans, respectively. During both seasons at Puñihuil the diet was almost

exclusively fish, with cephalopods and crustaceans combined accounting for $< 5\%$ of the mass.

At Pan de Azúcar cephalopods were eaten in larger quantities during the first sampling period (36%), but much less during the second sampling period (10%). Contribution to the diet by stomatopods in terms of wet mass was greatest during the first sampling period at Pan de Azúcar, but during both samplings crustaceans contributed only to a small degree on a mass per stomach basis.

Fish

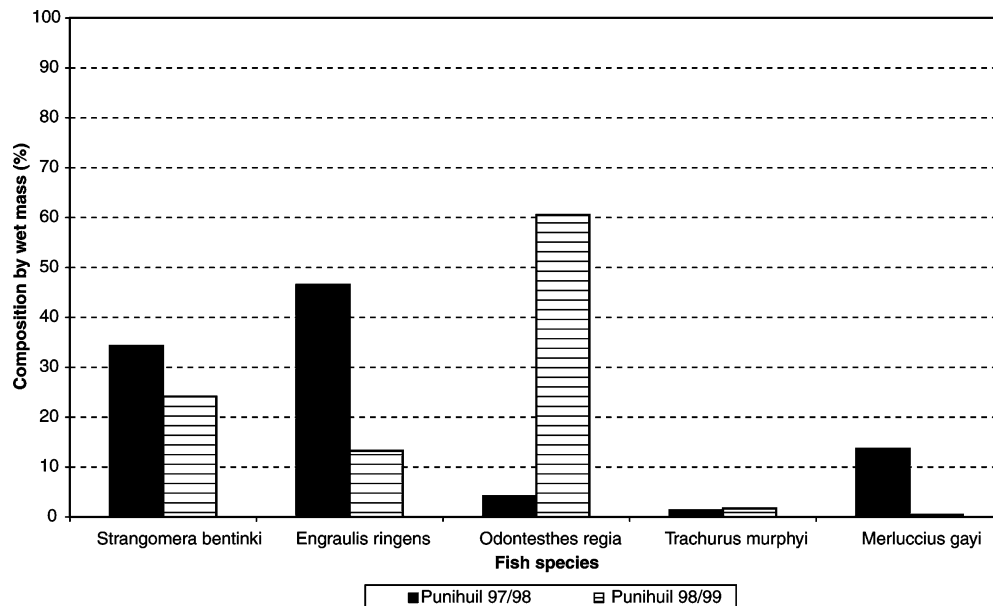
At Puñihuil most individual fish were relatively small, mean lengths ranging between 10.84 and 28.88 cm and mean masses ranging between 8.84 and 156.12 g (Table 4). Size of anchovy showed no significant differences between seasons ($P = 0.171$, Mann–Whitney U -test). At Puñihuil in 1997/1998 anchovy contributed most of the fish prey mass (47%), with silversides being more important (61%) than anchovies during the successive year (Fig. 3). Araucanian herrings were the second most important fish prey at Puñihuil during both seasons and showed no significant differences in size between seasons ($P = 0.065$, Mann–Whitney U -test).

At Pan de Azúcar, in all samples, mean lengths ranged between 5.4 and 15.06 cm and mean mass between 2.24 and 25.04 g (Table 4). The sizes of garfish eaten by Humboldt penguins during the first and second sampling periods were significantly different ($P < 0.000$,

Table 4 Mean length (cm) and mass (g) of the different fish species of in the stomach content of Humboldt penguins (*Spheniscus humboldti*) at Puñihuil (1997/1998 and 1998/1999 samplings) and Pan de Azúcar (all, first and second samplings)

	N	Length (cm)	SD	Weight (g)	SD
Puñihuil 1997/1998					
<i>Engraulis ringens</i>	111	11.09	1.93	8.84	4.46
<i>Strangomera bentincki</i>	65	11.04	1.28	12.04	3.30
<i>Odontesthes regia</i>	2	19.73		47.45	
<i>Merluccius gayi</i>	2	28.88		156.12	
<i>Trachurus murphyi</i>	1	14.7		30.41	
Puñihuil 1998/1999					
<i>Odontesthes regia</i>	48	19.52	3.58	49.73	30.13
<i>Strangomera bentincki</i>	41	10.97	1.72	11.19	6.95
<i>Engraulis ringens</i>	35	10.96	2.59	10.42	4.20
<i>Trachurus murphyi</i>	3	10.84	2.75	13.53	9.45
<i>Merluccius gayi</i>	1	13.51		11.19	
Pan de Azúcar, all samples					
<i>Scomberesox saurus</i>	218	15.06	3.99	7.08	7.73
<i>Engraulis ringens</i>	67	8.36	1.39	3.76	1.88
<i>Trachurus murphyi</i>	16	13.23	2.89	25.04	15.89
<i>Sardinops sagax</i>	7	5.40	1.18	2.24	1.1
Pan de Azúcar, first sampling					
<i>Scomberesox saurus</i>	30	19.25	5.60	15.17	12.10
<i>Engraulis ringens</i>	36	8.34	1.38	3.71	1.67
<i>Trachurus murphyi</i>	11	13.91	3.52	26.20	19.29
<i>Sardinops sagax</i>	2	4.10		1.04	
Pan de Azúcar, second sampling					
<i>Scomberesox saurus</i>	188	14.40	3.22	5.79	5.88
<i>Engraulis ringens</i>	31	8.37	1.44	3.82	2.13
<i>Trachurus murphyi</i>	5	13.32	0.43	22.51	2.20
<i>Sardinops sagax</i>	5	5.93	0.49	2.72	0.74

Fig. 3 Composition by wet mass (%) of the fish component of the diet of Humboldt penguins (*Spheniscus humboldti*) at Puñihuil in 1997/1998 and 1998/1999



Mann–Whitney *U*-test). Garfish predominated in the fish diet during both sampling periods in terms of wet mass, but the contribution of garfish to the fish prey was lower during the first sampling period (52%) than during the second sampling period (82%). Pilchards contributed only to a small degree to mass (on a per stomach basis) during both sampling periods, with 0.2% during the first sampling period and 1% during the second sampling period (Fig. 4).

Cephalopods

Most cephalopods were relatively small, with a mean length of 5.2 ± 1.4 cm and a mean mass of 5.5 ± 3.0 g at Puñihuil in the first year and a mean length of

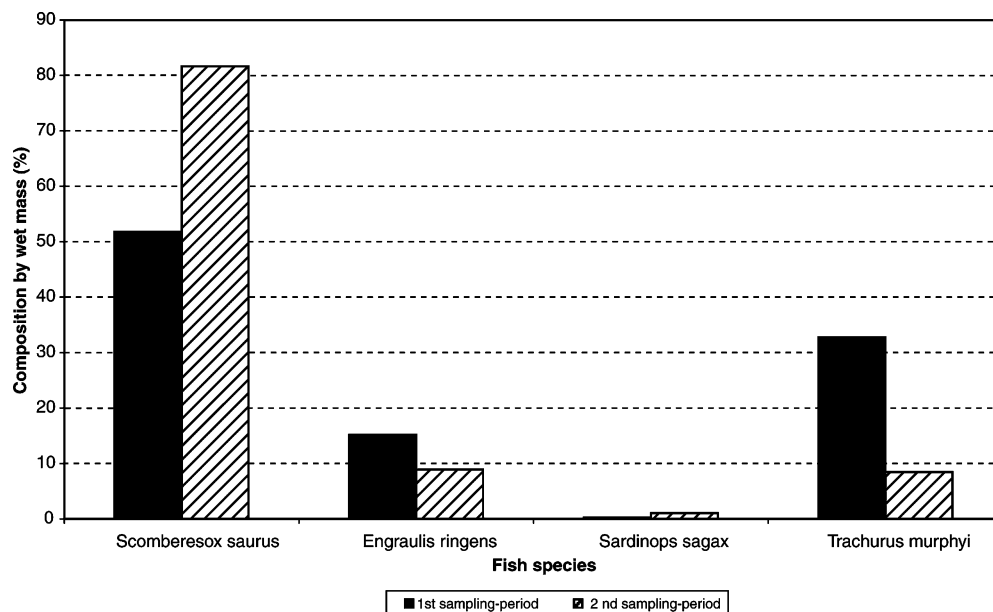
6.4 ± 1.4 cm and mean mass of 8.7 ± 4.3 g in the successive season. The length was significantly different between the two seasons ($P=0.000$, Mann–Whitney *U*-test). At Pan de Azúcar individual cephalopods were somewhat larger in size, with a mean length of 10 cm.

Discussion

Methodology

Problems associated with differential retention and digestion of diagnostic prey remains are inherent in any study of seabird diet. Digestion and passage rates of hard prey remains like otoliths or cephalopod beaks are

Fig. 4 Composition by wet mass (%) of the fish component of the diet of Humboldt penguins (*Spheniscus humboldti*) at Pan de Azúcar during the first sampling period and the second sampling period



subject to variation. This variation is affected by the relative size and thickness of the prey remains, their chemical composition and the acidity of the predator's stomach (Jobling and Breiby 1986). In general, the keratinous cephalopod beaks are more liable to retention than the calcareous otoliths. Accumulation of cephalopod beaks can result in overestimates of cephalopods (e.g. Van Heezik 1989). In the case of consumed fish, with small, fragile, hyaline otoliths, it is probable that some were digested completely and the numbers and their contribution to the diet were therefore underestimated. By using vertebrae for fish identification, we tried to avoid underestimation of the relative importance of fish, since the vertebrae showed little or no sign of digestion. Investigations on seals showed that the rate of identification is much higher when using otoliths as well as vertebrae (Pierce et al. 1991). Most cephalopod beaks present in the stomachs of Humboldt penguins (*Spheniscus humboldti*) in this study were still associated with flesh. This suggests that these prey items were caught recently, reducing the bias from accumulation.

Penguin stomachs in the breeding season 1997/1998 at Puñihuil were flushed up to five times, whereas in the following breeding season stomachs were flushed only twice. Since the aim of this study was not the comparison of the absolute numbers of prey items ingested, but the composition of the diet and the contribution of the different prey species to it, we only compared percentages. This enabled us to compare the different breeding seasons at Puñihuil despite the different flushing intensities.

Prey composition

In spite of the problems and biases which act in favour of underestimating the relative importance of fish and overestimating that of cephalopods, fish were still the dominant prey in Humboldt penguin stomachs. Humboldt penguins in this study took mainly anchovies (*Engraulis ringens*), Araucanian herring (*Strangomera bentincki*), silversides (*Odontesthes regia*) and garfish (*Scomberesox saurus*) (Figs. 1, 2). Contribution of the different species to the diet differed between sites and seasons. Penguins in the north eat mainly garfish and anchovies, which is consistent with the study on Humboldt penguins by Wilson et al. (1995), who found at Chañaral, in the north of Chile, that garfish was the dominant prey item and occurred in 94% of penguin stomach samples, anchovies being the second most abundant fish species. Of all items eaten, the proportion of garfish consumed in the present study was more than four times higher during the second sampling period than during the first (76% vs. 18%). Even though the contribution of anchovies to the total numbers of fish consumed was higher during the first sampling period, there were no significant differences between the proportion of anchovies of all items eaten during both

sampling periods. Thus, the low amount of garfish during the first sampling was compensated during the second by preying on cephalopods and crustaceans, mainly stomatopods. In northern Chile pilchards (*Sardinops sagax*) contributed only to a small degree to the penguin's diet, in this study as well in the study by Wilson et al. (1995).

At Puñihuil seasonal shifts between prey taxa occurred. In 1997/1998, penguins consumed almost exclusively anchovies, while they fed primarily on silversides in the successive year (Fig. 1). The second most abundant fish species preyed on at Puñihuil was in both years Araucanian herring. Silversides and Araucanian herring were absent in the study of Wilson et al. (1995). Instead, South American pilchard (*S. sagax*) was an important prey species, which, in turn, was absent in our investigations.

The principal prey species of this study are predominantly found in inshore areas (<http://www.fishbase.org>), which is consistent with the study by Culik and Luna-Jorquera (1997). They reported the foraging range of satellite-tracked Humboldt penguins to be between 2 and 92 km from Pan de Azúcar; 90% of all satellite locations came from a range of 35 km around the island; 50% came from a range of 5 km around the island.

Comparison with diet studies of congeners

The Humboldt penguin is similar to the other penguins of the genus *Spheniscus* in specialising on pelagic school fish. African penguins (*S. demersus*) have been reported to feed on anchovies (*Engraulis capensis*), pilchards (*Sardinops ocellata*) and pelagic gobies (*Sufflogobius barbatus*) (Crawford and Shelton 1981, cited in Sclaro et al. 1999; Randall and Randall 1986). Magellanic penguins (*S. magellanicus*) off Argentina take anchovies (*Engraulis anchoita*), sprats (*Sprattus fuegensis*), hake (*Merluccius hubbsi*), silversides (*Odontesthes* spp.) and squid (Frere et al. 1996; Sclaro et al. 1999). Sprats were the main food item of Magellanic penguins in southern Chile. Other prey were cephalopods like *Loligo* spp. or, rarely, other fish species (Wilson et al. 1995; Radl and Culik 1999). Galapagos penguins (*S. mendiculus*) were reported to feed on pilchard (*Sardinops sagax*) (Mills 2000). In previous investigations Humboldt penguins have been reported to feed on garfish (*Scomberesox* spp.), anchovy (*Engraulis ringens*), pilchard (*Sardinops sagax*) and squid (*Todarodes filippovae*) (Wilson et al. 1989, 1995).

The studies of Frere et al. (1996) and Sclaro et al. (1999) on Magellanic penguins off Argentina show clearly different feeding preferences over their breeding range. Penguins in the north feed mainly on anchovy (*Engraulis anchoita*), whereas penguins in the south of Argentina consume primarily sprat (*Sprattus fuegensis*) (Frere et al. 1996) and squid (Sclaro et al. 1999). African penguins show similar feeding preferences along their breeding range. African penguins breeding in the

north exclusively take pelagic gobies (*Sufflogobius barbatus*) (Crawford and Shelton 1981, cited in Sclaro et al. 1999); farther south the main prey is anchovy (*Engraulis capensis*) (Randall and Randall 1986). Previous studies on Humboldt penguins in Chile show similar results. In the north Humboldt penguins ate almost exclusively garfish; in central Chile they consumed anchovy, pilchard and squid; and in the south the penguins consumed anchovy and pilchard (Wilson et al. 1989, 1995). Differences in prey species taken by *Spheniscus* penguins between sites in the studies mentioned above, as well as in our study, seem to be a reflection of the availability of pelagic school fish in the foraging region. Already Wilson et al. (1995) and Sclaro et al. (1999) noted that inter-site variability is greater than inter-specific differences in prey species taken by *Spheniscus* penguins.

Seasonality

Some fish species show seasonal variation in their distribution (e.g. Parrish et al. 1989; Castillo et al. 1996). Even in regions with frequent and strong upwelling events like in northern Chile, fish species like pilchards (*Sardinops* spp.) show migratory behaviour between the winter spawning season and summer and fall (Parrish et al. 1989). Seasonal differences in the abundance of prey species lead to variation in their availability throughout the year. Seasonality in the diet of *Spheniscus* penguins has already been documented (e.g. Wilson 1985; Sclaro et al. 1999).

The high abundance of cephalopods and crustaceans during the first sampling period at Pan de Azúcar may, therefore, indicate seasonal differences in the diet of Humboldt penguins and a change of prey species during the breeding season. Similar results were found by Sclaro et al. (1999). Magellanic penguins in Argentina, at Monte Leon, fed almost exclusively on cephalopods in November, whereas in December cephalopods only accounted for about 30% of all the prey items consumed. This shows that the prey composition of the penguins can change in short periods of time.

In 1997/1998, penguins at Puñihuil consumed almost exclusively anchovy, while they fed primarily on silversides (*Odontesthes regia*) in the successive year (Fig. 1). Fish of the genus *Odontesthes* have already been documented to occur in the diet of Magellanic penguins (e.g. Sclaro et al. 1999). The seasonal differences in the diet of the Humboldt penguin may reflect a change in the availability of fish species during the season, as was also documented for Magellanic penguins off Argentina (Sclaro et al. 1999). Magellanic penguins consumed squid almost exclusively for most of the breeding season, but silversides were the major prey item in December (<60%). Because our investigations during the first season were conducted in February, but in March during the successive season, the differences we found may reflect seasonal as well as annual variability.

Oceanographic conditions

Previous studies have shown that physical oceanographic conditions like sea surface temperature and salinity have an influence on distribution and availability of marine resources (e.g. Castillo et al. 1996; Hansen et al. 2001). Differences in the oceanographic conditions along the coast off Chile, resulting from climatic conditions like wind, insolation and rainfall, which in addition vary seasonally, seem to lead to different spatial fish distributions. The relationship between physical oceanographic conditions and pelagic fish distributions off northern Chile has been studied by Castillo et al. (1996); they showed that the distribution of three pelagic fish species was associated with the occurrence and intensity of thermal and haline fronts. Annual and regional differences in the prey species taken in our study may therefore also be related to different thermal fronts during the sampling periods.

In the case of El Niño events, when oceanographic conditions change and sea surface temperatures increase, fish stocks have been reported to migrate to cold-water areas (e.g. Parrish et al. 1989; Arntz and Fahrbach 1991). The impacts of poor fish availability on penguins during El Niño events have been widely studied (e.g. Hays 1986; Duffy et al. 1987b; Boersma 1998). The consumption of stomatopods at Pan de Azúcar can possibly be related to the El Niño phenomenon in 1997/1998, which extended late into 1998. Stomatopods have never previously been recorded in the diet of Humboldt penguins. These benthic crustaceans seem to be atypical prey for these seabirds and may reflect poor availability of fish in the beginning of the post-El Niño breeding season. Duffy et al. (1987b) observed a shift from fish to cephalopod prey in Magellanic penguins as a result of an El Niño event. It might be that the oceanographic conditions at our study location changed between the two sampling periods, as a La Niña event was expected at the end of 1998 (<http://www.wmo.ch/nino/updat.html#intro>). However, since no detailed information on the oceanographic conditions was collected in the current study, this explanation has to remain speculative.

Commercial fisheries

The commercial fishery of Chile intensively exploits of the main prey species of Humboldt penguins, e.g. anchovy, silversides, garfish and Araucanian herring. In 1997, the first year of our study, the total catch of these species was 2.2 million tons (Sernap 1998). This represents 34% of the overall catch of the entire Chilean industrial fishery in the same year.

Anchovy is a commercially harvested species (Whitehead et al. 1988), with Chilean landings of $>0.5 \times 10^6$ t in 1998 (Sernap 1999). In the region in the north of Chile, where Pan de Azúcar is located (region III), the total catch of anchovy was 54,847 t, with peaks of landings from January to August (Sernap

1999). Even when the landings of anchovy decreased towards the end of the year, anchovies were the most harvested fish species in this region. Because schools of garfish are smaller than those of anchovies (<http://www.fishbase.org>), garfish are only to a small degree of interest to the commercial fisheries (Sernap 1998, 1999). There has been no report on the commercial exploitation of garfish in region III in the north of Chile (Sernap 1999). The predominance of garfish in the north might indicate that the birds could switch from one prey species to another if overfishing of anchovies occurs.

At Puñihuil the abundance of anchovy in the penguin stomach contents was high in the first season and lower in the second. The total catch of anchovy in the years and months of the study show the opposite results. Landing of anchovy in region X of Chile, where our study area was located, were low in 1998, with no landings reported from January to April, whereas landings of anchovy added up to >1,800 tons in the same months of the successive year (Sernap 1999, 2000). The lower amount of anchovy in the penguins' diet in March 1999 at Puñihuil may be related to the high numbers of anchovy landings, especially in the first 4 months of the year. The total catch of silversides was different between years, with total landings in region X of 416 tons in 1998 and 3,271 tons in 1999 (Sernap 1999, 2000). During 1998 the total landings of silversides were 50 tons or less in 9 months and >50 tons in 3 months of the year. These results suggest that the abundance of silversides was low during that year, but was higher in the successive year, which is shown in the greater amounts of silversides in the penguins' diet and the nets of the fisheries in 1999.

Food requirements during foraging

Knowledge of diet composition is an important basis for understanding key parameters of seabird energetics and breeding performance (Furness and Monaghan 1987). Because, e.g., cephalopods have a lower energy content than fish, the greater the proportion in the diet, the lower the energy per stomach load. Studies have shown that African penguins digest cephalopods only half as fast as anchovy (Wilson et al. 1985) and that chicks grow more slowly when fed with squid compared to anchovy (Heath and Randall 1985). Duffy et al. (1987b) connected the massive mortality of Magellanic penguins during El Niño events with a change in the prey spectrum, which shifted from anchovies to cephalopods. Knowledge on the composition of the diet is therefore important to estimate potential difficulties for the penguins in meeting their energetic demands.

Energy requirements of adult Humboldt penguins at sea and ashore have been investigated earlier (e.g. Luna-Jorquera 1996b; Culik and Luna-Jorquera 1997; Culik et al. 1998; Luna-Jorquera and Culik 2000). Luna-Jorquera (1996b) calculated that the energetic requirements of Humboldt penguins for the 70 days of the

chick-rearing period, including time at sea and ashore, are 2,240 kJ on average per day per individual penguin.

Assuming a diet almost exclusively of fish (on a wet mass basis), as found at Puñihuil, which has a higher energy density than a diet containing cephalopods or crustaceans and which resulted in higher reproductive success (Hennicke, unpublished data), a mean energy content of 5.93 kJ g⁻¹ wet mass (mean energy content for anchovy, Fitzpatrick et al. 1988) and an assimilation efficiency of 77.3% (Guerra 1992), an individual breeding penguin would need to consume 489 g day⁻¹ to balance its mean energy requirements. For the relatively small colony at Puñihuil of about 150 Humboldt penguins, the fish consumed during the 70 days of breeding add up to 5.1 tons. If these demands exceed the food availability in adjacent waters, they may lead to breeding failure and consequently to population decline.

The total population of Humboldt penguins in Chile is estimated at 40,000 individuals (Bernal, personal communication). The energetic demands of the total population during the breeding season sum up to 1,400 tons of fish. The intensive commercial exploitation of the main prey species of the Humboldt penguin shows that the commercial fishery is potentially capable of harming the penguin population if overfishing occurs. Because the potential threat to Humboldt penguins is serious, continued monitoring of feeding preferences and diet composition could do much to alert authorities to changes in the marine ecosystem on which these birds depend and to help improve fisheries management to avoid undue competition between man and penguins.

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